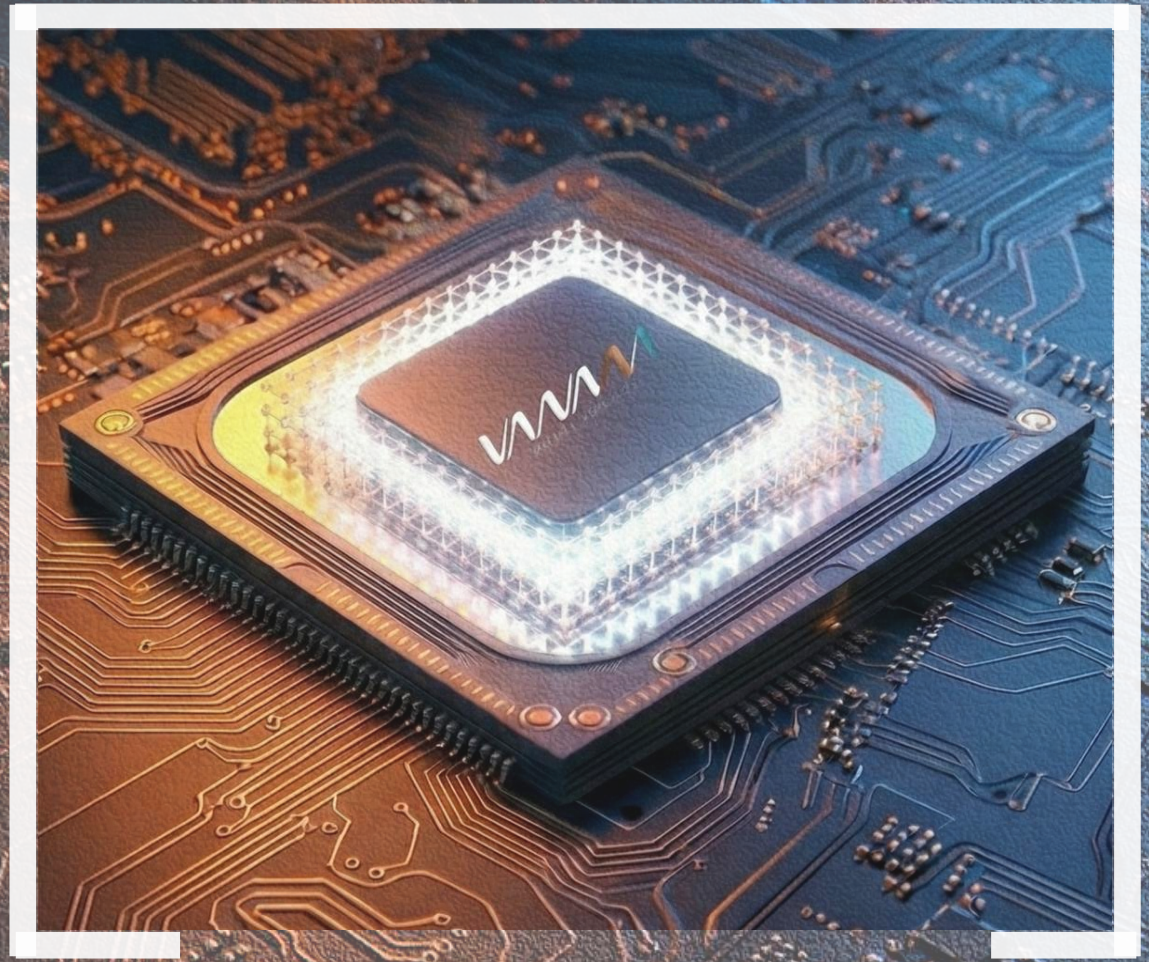


Next-Generation Foundry for Quantum Photonics

VANA MATERIALS LAB.

Company Overview

> Founded in 2022, dense micro-foundry foundation in Seoul, current strategic focus on quantum photonics.

COMPANY FACTS

Founded	Jan. 27, 2022
1st Lab	Yeongdeungpo, Seoul
Fab footprint	~1,600 ft²
Team	13 employees
Management	PhD-led
Customers	42 paying
IP	26 patents + 3 TMs
Funding	Seed & $3M+ non-dilutive

OPERATING CAPABILITY

1. **Deposition**
   Cluster Sputter/ Inline Sputter/ ALD / Evaporator
2. **Metrology**
   Ellipsometer / Parameter analyzer
3. **Back-end**
   Laser dicing; external packaging and cryogenic test network
4. **Materials**
   50+ oxides, nitrides and metals; quantum focus on NbTiN, Ta_2O5, BTO, STO and DBR stacks

STRATEGIC EVOLUTION

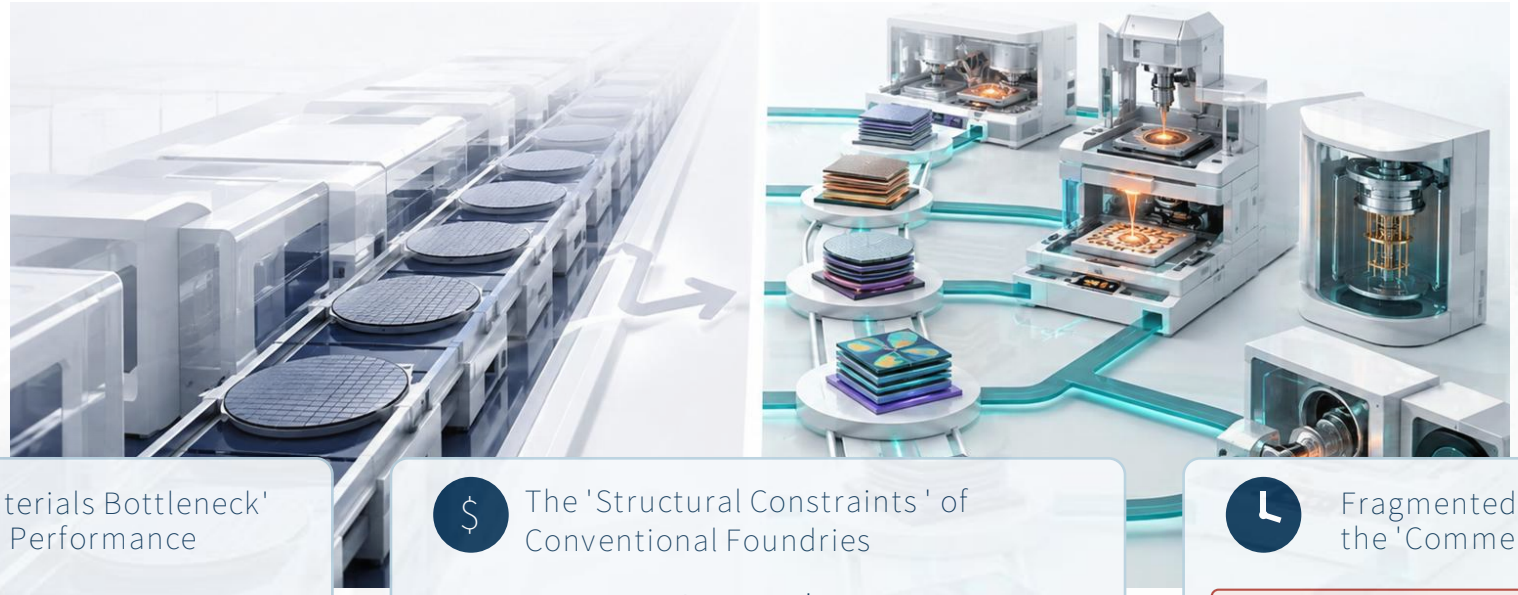
- **2022** — Company founded; Seoul micro-foundry initiated
- **2023-25** — Seed funding, R&D grants, customer projects, portfolio and IP build-out
- **2025-26** — U.S. market exposure through Luminate Accelerator and U.S. events
- **2026-28** — **Focused build of U.S. quantum photonics foundry**

NEW POSITIONING: SPECIALTY QUANTUM PHOTONICS FOUNDRY

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Problem

> Thin Film Material Gaps · Structural Foundry Constraints · Fragmented Supply Chain



The 'Advanced-Materials Bottleneck' Limiting Quantum Performance

$10^{-4} \rightarrow 10^{-9}$ /K

Si thermo-optic coefficient drops by 10^5 from 300 K to 4 K, quantum optical control is lost

31 → 24 pm/V

Standard $LiNbO_3$ Pockels coefficient drops ~20% at cryogenic temperature, with added heat load

Komma et al., APL 2012 · Herzog et al., Opt. Commun. 2008 · Pintus et al., Nat. Electron. 2022

The 'Structural Constraints ' of Conventional Foundries

4–6 months · $500K+

One standard fab cycle (fixed material stack only)
MPW (Multi-Project Wafer): $50K–$100K
Custom process setup + masks (NRE): $2M–$5M

Early-stage quantum MOQ	Conventional foundry MOQ
5–50 wafers	1,000+ wafers
Small-lot	Mass-production MOQ

U.S. DoD ManTech · AIM Photonics (cycle) · NRE = cycle × iterations (arithmetic) · 5–50 wafers: VanaM customer interviews

Fragmented Supply Chain: the 'Commercialization Gap'

4+ Supply Chain

A Deposition → B Patterning → C Packaging → D Test

Yield loss: interface oxidation and contamination during transport / queue exposure
Lead-time delay: one standard fab cycle takes 4–6+ mon

Cycle 4–6 months · $500K+: U.S. DoD ManTech · Supply-chain stages: VanaM customer cases (internal) · CNAS 2026

Needed: a " Specialty Quantum Photonics Foundry" — materials development, patterning, and cryogenic validation in one workflow

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Customer Pain Points

> Time & capital loss for early quantum hardware teams due to vendor-fragmented deposition, nano-patterning, metrology & cryogenic validation.

MATERIAL ACCESS

"We cannot source the film at the required stoichiometry and thickness."

PROCESS FLEXIBILITY

"The foundry will not alter its standard process stack."

LOT ECONOMICS

"We need 5-50 wafers, not a volume contract."

VENDOR HANDOFF

"Deposition and nano-patterning sit with different vendors."

LATE VALIDATION

"Cryogenic failure is discovered after the fabrication cycle."

IP / DATA

"Recipe data and process IP ownership are unclear."

Impact: longer qualification cycles | duplicated setup cost | uncertain yield | diversion of scarce quantum engineering talent

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Solution: VanaM Specialty Quantum Photonics Foundry

> Patented, AI-assisted process platform for next-generation quantum photonics commercialization.



- Custom quantum photonics materials & device iteration.
- AI-assisted process for oxides, nitrides, metals & multilayers.
- Low-volume, high-value & rapid-turnaround dedicated lots.
- Recipe, process-window & device co-development options.
- Custom nanowire/photonic structures within tool capacity.
- DOE-based recipe & device co-development.
- Partnered cryogenic test & package path.



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Solution: Quantum Photonics Foundry-Platform

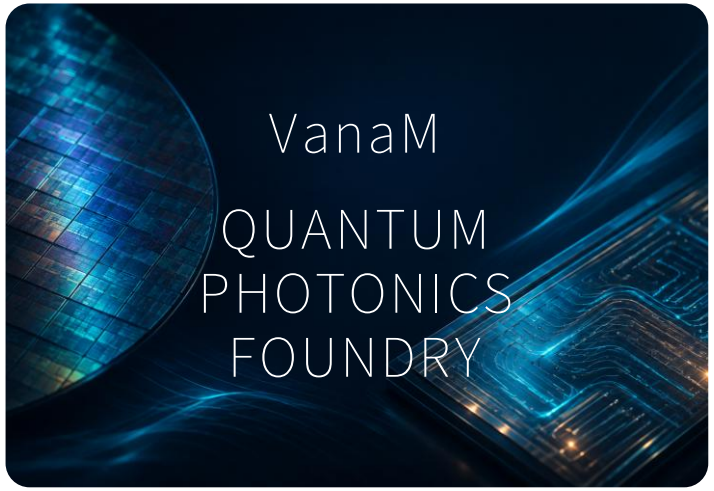
> VanaM: one commercial workflow for custom material deposition + nano-patterning device fabrication + qualification data + partner-enabled packaging.

1 Materials Foundry

Custom oxides, nitrides, metals and multilayer stacks; recipe development and low-volume wafers.

2 Device Foundry

Nano-patterned SNSPD, waveguide, modulator, DBR and PIC structures within qualified process windows.



3 Process Qualification

DOE, metrology, run-to-run data, specification control and recipe transfer from Korea to the U.S.

4 Integration Network

Dicing, packaging, fiber coupling and cryogenic test through internal capability and qualified partners.

NRE / DOE → Pilot Wafer → Patterned Device → Repeat Lot → Package / Test → IP / JD

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Business Model

> Value before scale-up, then wallet expansion as customers move from samples to wafers, devices, packaged components & PIC supply.

1 **Process NRE**
DOE, recipe setup, engineering and qualification milestones

2 **Material Wafers**
Korea-made thin-film wafers: NbTiN, Ta2O5, BTO/STO and DBR stacks.

3 **Patterned Devices**
U.S. lithography, nano-patterning, etch, release and screened die sales.

4 **Package / Test**
Dicing, fiber/electrical package and cryogenic performance verification.

5 **Integrated Quantum PIC**
1Q-eq evaluation PIC, 8Q module and 32Q tile supply programs.

6 **IP / Joint Development**
License, royalty, exclusivity or co-development around platform processes.

CUSTOMER MATURATION AND WALLET EXPANSION

Sample / Feasibility
$10k-$50k
Material sample, characterization or limited DOE.

>

Paid Qualification
$50k-$250k
Process development and pilot wafers.

>

Material / Device Supply
$0.3M-$1.5M
Recurring wafers, dies, packages and engineering support.

>

Integrated PIC Eval Lot
$0.5M-$2.0M
10-50 1Q-eq PICs, 8Q modules and qualification support.

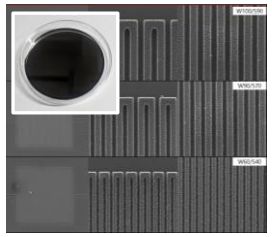
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Strategic PIC Supply
$2M-$10M+
Multi-tile supply, platform JD, exclusivity or annual framework.

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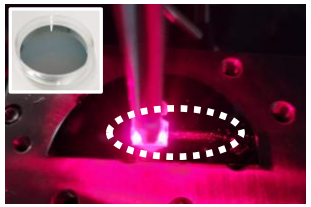
Quantum Photonics Product Portfolio

SNSPD



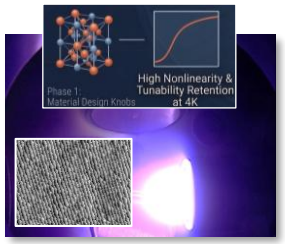
High-sensitivity superconductors

Waveguide



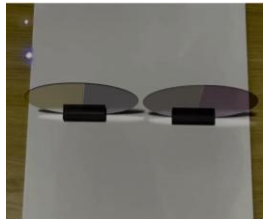
Low loss optical waveguides

Modulator



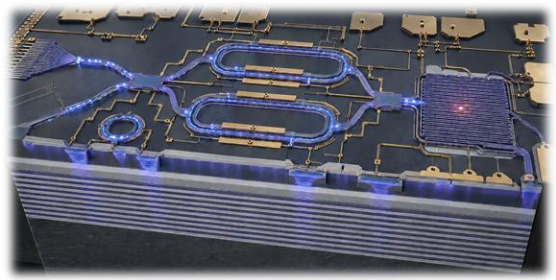
High cryogenic electro-optic modulation

DBR



High-reflection DBR wafer

Photonics Integrated Circuits (PIC)



Key Thin Films Next-Gen PICs & for Quantum Systems

- ✓ SNSPD: NbTiN ⟶ Superconducting transition T ~ 11K
- ✓ Waveguide: Ta_2O_5 ⟶ Propagation loss < 0.5 dB
- ✓ Modulator: BTO, STO ⟶ Pokel coefficient ~1000pm/V at 5K
- ✓ DBR: Si-SiO_2-TiO_2 ⟶ Reflectance > 99%

KIST Quantum Fab & UIUC Collaboration

Product Ladder

1 Thin film materials wafer
Korea Foundry
Certified NbTiN, Ta2O5, BTO/STO, DBR multilayer

2 Patterned Device/Die
Korea + U.S. Foundry
SNSPD die & Chip, Waveguide, Modulator, Patterned DBR

3 Quantum Photonics PIC
U.S Foundry
$30K ASP; Max 300ea/Mon = $108M/yr

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Market Expansion Opportunities

> Bottom-up account investable market case & wallet-share opportunity within a much larger, fast-growing quantum and advanced photonics ecosystem.

MARKET SIZE VALUE

$60B By 2030	TAM	Quantum computing, communication, sensing, silicon photonics, LiDAR, advanced photonics
$4B By 2030	SAM	Custom quantum photonics thin films, wafers, devices and related process services
$50M Current market, Bottom-up	BEACHHEAD	U.S. low-volume, high-value customers requiring custom wafers and nano-patterned devices

Growth signal — *McKinsey: quantum communication ~22-25% CAGR through 2035; core quantum technology revenue up to $97B by 2035.*



Sources: Grand View Research, MarketsandMarkets, Yole Group, KBV Research, 2025–2026 industry reports.

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Traction & Validation

> U.S. market entry with operating KR fab, growing repeat-customer revenue base, 50+ material process library, non-dilutive funding & recognized U.S. photonics ecosystem access.

KOREA CUSTOMER BASE QUALITY (FY ~26)

42 Active accounts
cumulative 42 paying customers to date

Active accounts	42
Repeat accounts (2+ orders)	29
Repeat-order share of revenue	91.3%
Average annual value per Repeat accounts	$8.6K
'23 ~ '25 Active accounts CAGR	169.3%
University : R8D Institute : Corporate	55 : 25 : 20

MATERIALS & PROCESS LIBRARY (49 FILMS)

50+ Thin-film materials — process secured
special films & process windows beyond conventional fabs



ECOSYSTEM ANCHOR POINTS (public)

KIST Quantum Fab · UIUC
Collaboration pathways for cryogenic and device validation

Intelon Optics (Boston, MA)
Spectrometer PIC & CHIPS act grant proposal partner

Stratio Inc. (San Jose, CA)
Photonics-sensor ecosystem partner

NextCorps Luminate Cohort 8 (Rochester, NY)
AIM Photonics · Thorlabs · Edmund Optics · Omega Optical network touchpoints

SelectUSA Investment Summit 2026 — 1st place, Open Tech Pitching (2025: 2nd)
Repeat top ranking before U.S. investors and EDOs

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U.S. Expansion – Strategy

> A new U.S entity to build and operate Phase 1 Foundry, KR-US dual-site model.

KOREA FOUNDRY

PROCESS DEPTH + REVENUE

- Existing customer production
- Material recipe development
- Cross-site transfer standards
- Metrology and technical support
- Training and troubleshooting

U.S. PHASE 1 FOUNDRY

8-INCH + NANO-PATTERNING

- Customer-facing process qualification
- 8-inch deposition and multilayer stacks
- Nano-patterned quantum photonic devices
- U.S. technical team and program access
- Domestic supply-chain positioning

PARTNER NETWORK

EXTEND THE VALUE CHAIN

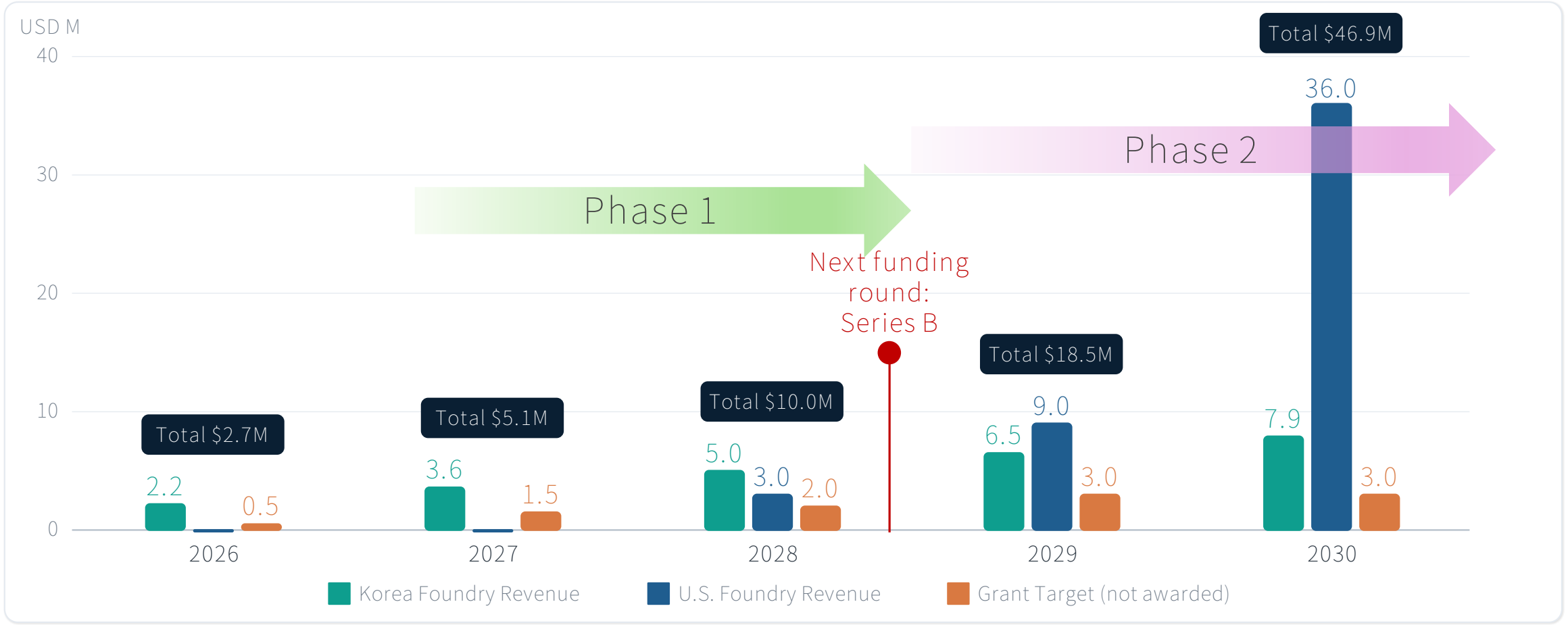
- Packaging and fiber coupling
- Cryogenic device test
- Advanced SEM / AFM / optical metrology
- University and national-lab programs
- Scale-foundry and strategic partners

Phase 1: customer qualification platform (2026-2028)
Phase 2: automated 8-inch quantum photonics foundry (post-2028, additional capital)

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Financial Projections

Model growth from $2.7M in 2026 to $46.9M in 2030, U.S. foundry as main growth driver after 2028.



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Vision & Ask

> Funds focused on U.S. Phase 1 to convert KR-developed materials & process IP into 8-inch U.S. manufacturing platform.

Target:
$8M*
Series A
Pre-money: 15-20%**

Use of Funds: U.S. Foundry

- Foundry Phase 1
- 8-inch + nano-patterning → PICs
- Team Expansion
- International Events & Marketing

- * + Projected $2M USD/year as non-dilutive funding (Korea & US R&D grants)
- ** Open to discussion

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Management Team

> PhD materials leadership, thin-film deposition research team, IP & cleanroom construction expertise



Dr. Soodeok Han

CEO / Founder

PhD in Engineering; semiconductor thin films; commercialization and foundry strategy.



Dr. Wonyoung Choi

CTO

PhD in Engineering; semiconductor thin films; process and R&D leadership.



Dr. Sahar Ayachi

CSO

PhD in Engineering; semiconductor devices; international technical programs.



Dr. Hyeonkyo Song

Principal Researcher

PhD in Physics; semiconductor devices and advanced materials research.

PLANNED NEXT-STAGE HIRE

$

Finance Lead

- U.S. subsidiary finance
- Grant accounting
- Capex control
- Investor reporting
- Follow-on fundraising

OPERATING CAPABILITIES

13	PhD-led	Deposition	IP	Cleanroom
Employees (full-time)	Founding & Management team	Specialist research staff	Patent attorney expertise	Infrastructure architect

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| Contact

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